

JACOB SHKROB

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Education

New York University Ph.D. Mathematics	2022 – 2027 (expected)
McGill University B.Sc. Honors Probability & Statistics, Minor in Computer Science	2018 – 2022 <i>CGPA: 3.92/4.00</i>

Honors & Awards

Charles Newman Fellowship <i>New York University — awarded to an outstanding graduate student in probability</i>	2025
Finalist, Azrieli Visiting Ph.D. Fellowship <i>Azrieli Foundation</i>	2026
1st Place & \$25,000, ProjectX Undergraduate AI Research Competition <i>Clinical practice cohort — personalizing mechanical ventilation via deep conservative reinforcement learning</i>	2022
Faculty of Science Merit Scholarship <i>McGill University — awarded in both 2019–2020 and 2020–2021</i>	2019 – 2021
Dean's Honors List <i>McGill University — awarded in both 2019–2020 and 2020–2021</i>	2019 – 2021
Winner, PharmaHacks (Novartis clinical trial data track) <i>Montreal Institute for Learning Algorithms (MILA)</i>	2019

Publications & Preprints

- [1] J. Niles-Weed and **J. Shkrob**. Nearly sharp comparison results for sliced and max-sliced Wasserstein distances. *arXiv:2608.13374*, 2026.
- [2] Y. Han, A. Shetty, and **J. Shkrob**. An empirical Bayes perspective on heteroskedastic mean estimation. *arXiv:2603.13499*, 2026.
- [3] J. Niles-Weed, S. Sadosky, and **J. Shkrob**. Vector balancing with permutations. In preparation, 2026.
- [4] J. Niles-Weed and **J. Shkrob**. Bayes-optimal estimation for Gaussian planted matchings. In preparation, 2026.
- [5] F. Kondrup, T. Jiralerspong, E. Lau, N. de Lara, **J. Shkrob**, M. D. Tran, D. Precup, and S. Basu. Towards safe mechanical ventilation treatment using deep offline reinforcement learning. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 37, pages 15696–15702, 2023.

Research Experience

AstraZeneca Student Researcher ↔ Dr. Natasha Markuzon, Dr. Damian Bikiel Simulated time-to-event data exhibiting both predictive and prognostic behavior. Developed methods for inducing subgroup structure.	May 2022 – Aug 2022
McGill Institute for Computational Medicine (MICM) Summer Research Intern ↔ Prof. Yi Yang - Developed an algorithm for the time-varying Cox proportional hazards model, with applications to survival analysis and immunology. - Wrote R packages in C++/R using Rcpp, adding parallelism to cut running time.	May 2020 – Nov 2020 <i>McGill University</i>
Big Data Summer Institute (REU) Student Researcher ↔ Prof. Bhramar Mukherjee Applied data mining and machine learning to genomic and phenotypic data with the goal of improving drug efficacy for individual cancer patients.	June 2019 – Aug 2019 <i>University of Michigan</i>

Conferences, Workshops & Summer Schools

Princeton Machine Learning Theory Summer School <i>Participant/Presenter</i> ↪ Princeton University • Doctoral training in theoretical machine learning, training dynamics, and diffusion models.	August 2026
Conference of Learning Theory <i>Participant</i> ↪ University of California San Diego • Presented work on empirical Bayes for heteroskedastic mean estimation.	June 2026
Workshop on Robust Statistics <i>Presenter</i> ↪ Columbia University • Presented work on empirical Bayes for heteroskedastic mean estimation.	April 2026
Statistical Optimal Transport Summer School <i>Participant</i> ↪ Simons Laufer Mathematical Sciences Institute • Doctoral training in statistical optimal transport methodology.	June 2025
Methods in Single-Cell Data Integration and Optimal Transport <i>Participant</i> ↪ Irving Institute for Cancer Dynamics • Multi-modal single-cell integration; emerging spatial and imaging technologies.	June 2025
Statistical and Numerical Methods for Non-commutative Optimal Transport <i>Participant</i> ↪ Institute for Pure & Applied Mathematics (UCLA) • Numerical algorithms for optimal transport and gradient flows.	May 2025
Emerging Directions Workshop, NITMB <i>Participant</i> ↪ University of Chicago • Discussed open problems in mathematical biology presented by leading researchers.	Feb 2025
Probability, Algorithms and Inference <i>Participant</i> ↪ Georgia Institute of Technology • Combinatorics of the perceptron model, planted subgraph recovery, sphere packing.	Summer 2024
51st Probability Summer School, Saint-Flour <i>Participant</i> • Bayesian nonparametric statistics, stochastic homogenization, integrable systems.	Summer 2023

Projects & Extracurriculars

PharmaHacks — Novartis Clinical Trial Data <i>Winning Project</i> ↪ Montreal Institute for Learning Algorithms (MILA) • Built random forest, naïve Bayes and XGBoost classifiers in <code>scikit-learn</code> and <code>pandas</code>. • Used χ^2 tests of independence, ANOVA and bagging/boosting for feature and model selection.	Fall 2019
Directed Reading Project <i>Undergraduate Project</i> • <i>Probabilistic analysis of simple random graph models.</i>	Fall 2020

Technical Skills

Programming	Python, R, C++, Java, MATLAB, OCaml, Bash (<i>learning Julia</i>)
Tools	L ^A T _E X, PyTorch, tidyverse, ggplot2, R Markdown, Git/GitHub, Vim, Anaconda, Excel
Methods	Statistical modeling, data wrangling and cleaning, survival analysis, ML pipelines
Languages	English, Russian, French, Hebrew (<i>basic</i>)